Hints: Number theory – 2

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Note. The Fermat-number claim is false for $k = 0, 1$ (3 and 5 are not $\equiv 1 \pmod 4$, resp. $\pmod 8$). The hints prove it for $k \geqslant 2$.

Hint 1. Theorem 1: write $m = q \operatorname{ord}_n(a) + r$ with $0 \leqslant r < \operatorname{ord}_n(a)$; then $a^r \equiv 1$, so $r = 0$. If $p \mid 2^p - 1$ for a prime p, then $\operatorname{ord}_q(2)$ (for a prime $q \mid 2^p - 1$) divides p and is not 1, so it equals p. Hence $p \mid q - 1$, and q is odd.

Hint 2. Let $q \mid F_k = 2^{2^k} + 1$. Then $2^{2^{k+1}} \equiv 1$ but $2^{2^k} \equiv -1$, so $\operatorname{ord}_q(2) = 2^{k+1}$. To gain one more factor of 2, find a square root of 2 mod q: for $k \geqslant 2$, check that $s = 2^{2^{k-2}}(2^{2^{k-1}} - 1)$ satisfies $s^2 \equiv 2 \pmod q$.

Hint 3. Then $\operatorname{ord}_q(s) = 2^{k+2}$, which divides $q - 1$. For primes $\equiv 1 \pmod n$: if $q \mid \Phi_n(a)$ and $q \nmid n$, then $\operatorname{ord}_q(a) = n$, so $q \equiv 1 \pmod n$. Apply this to $a = N!\, n$, where $\Phi_n(a) \equiv \Phi_n(0) = \pm 1 \pmod{N!\, n}$, so all its prime factors exceed N.

Problem 2

Hint 1. For $d \mid p - 1$ let $\psi(d)$ be the number of elements of order d. If $\psi(d) > 0$, these elements are among the d roots of $x^d = 1$, which form $\langle g \rangle$ for any one of them. So $\psi(d) \leqslant \varphi(d)$, and $\sum_d \psi(d) = p - 1 = \sum_d \varphi(d)$ forces equality.

Hint 2. Prime powers: if g is a primitive root mod p, then g or $g + p$ has order $p(p-1)$ mod p^2, and such an element stays primitive mod every p^k (use LTE on $g^{p-1} - 1$). For $2p^k$ take an odd primitive root mod p^k. For other n, CRT gives a product of two groups of even order, or $(\mathbb{Z}/2^k)^*$ with $k \geqslant 3$, and these are not cyclic.

Hint 3. $\operatorname{ord}_9(2) = 6$, and $v_3(2^{2 \cdot 3^j} - 1) = v_3(4 - 1) + j = 1 + j$, so $\operatorname{ord}_{3^k}(2) = 2 \cdot 3^{k-1} = \varphi(3^k)$. For $k \geqslant 2$, $2^j - 1 < 2^k - 1$ when $0 < j < k$, so the order of 2 modulo $2^k - 1$ is exactly k.

Problem 3

Note. As printed the problem is trivial: $1, 2, \ldots, p - 1$ is wonky for every $p > 3$, since $i(i+2) - (i+1)^2 = -1$ and both wrap-around values equal -3. The original (ICMC) also requires that the partial sums $a_1a_2,\ a_1a_2 + a_2a_3,\ \ldots,\ a_1a_2 + \cdots + a_{p-1}a_1$ be distinct mod p. The hints follow that version. The answer is yes for every $p > 3$.

Hint 1. Make the partial sums telescope: try to choose a_i so that $a_i a_{i+1} = c_i - c_{i+1}$ for a simple injective sequence c_i.

Hint 2. Take $a_i \equiv i^{-1} \pmod p$. Then $a_i a_{i+1} = \frac{1}{i} - \frac{1}{i+1}$, so the partial sums are $1 - \frac{1}{j+1}$ for $j \leqslant p - 2$. These are distinct and avoid 0 and 1.

Hint 3. The last sum adds $a_{p-1}a_1 = -1$ to $S_{p-2} = 2$, giving 1, which is new. Wonkiness: $a_i a_{i+2} - a_{i+1}^2 = \frac{1}{i(i+1)^2(i+2)} \neq 0$, and the two wrap-around values equal $-\frac{3}{2} \neq 0$ for $p > 3$.

Problem 4

Hint 1. The right-hand side counts ordered bases of $\mathbb{F}_p^n$: choose the i-th column outside the span of the previous ones, in $p^n - p^{i-1}$ ways. So it is $|\mathrm{GL}_n(\mathbb{F}_p)|$.

Hint 2. Find a subgroup of order $(p-1)^n n!$: the monomial matrices (permutation matrices times invertible diagonal matrices). Then apply Lagrange's theorem.

Problem 5

Hint 1. The b_n form an arithmetic progression $b_0 + nr$ of length $p-4$, so $1, a, \ldots, a^{p-5}$ are distinct, and $\mathrm{ord}_p(a) \geqslant p-4$. For $p > 7$ this forces $\mathrm{ord}_p(a) = p-1$.

Hint 2. The complement of an AP of length $p-4$ in $\mathbb{F}_p$ is an AP of length 4 with the same difference. The complement of $\{1, a, \ldots, a^{p-5}\}$ is $\{0, u, u^2, u^3\}$ with $u = a^{-1}$. Up to reversal it equals $\{0, d, 2d, 3d\}$ or $\{-d, 0, d, 2d\}$.

Hint 3. Write $u = \alpha d$, $u^2 = \beta d$, $u^3 = \gamma d$. Then $\beta^2 \equiv \alpha\gamma$, where $\{\alpha, \beta, \gamma\} = \{1, 2, 3\}$ or $\{-1, 1, 2\}$. Check the few cases: only $9 \equiv 2$, i.e. $p = 7$, survives with $p > 5$. For $p = 7$, $a = 3$ or 5 works. Answer: $p = 7$.